CASE PRESENTATION

PREDICTING HEART DISEASE, DIABETES, AND STROKE USING CLINICAL DATA

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Dataset

https://www.kaggle.com/datasets/rafi003/healthcare-disease-prediction-dataset/data?select=healthcare_disease_prediction_2000.csv

healthcare_disease_prediction_2000

Patient_ID	Age	Gender	BMI	Blood_Pressure_Systolic	Blood_Pressure_Diastolic	Cholesterol	Glucose_Level	Smoking	Alcohol_Intake	Physical_Activity	Family_History	Heart_Disease	Diabetes	Stroke
1	69	Male	34.1	127	71	184	154	1	1	0	0	0	1	0
2	32	Male	25.0	148	69	177	136	1	1	1	0	0	0	0
3	78	Male	25.2	98	93	252	110	0	0	0	0	0	0	0
4	38	Male	9.6	162	102	274	87	1	0	1	0	0	0	0
5	41	Female	29.0	161	69	251	127	0	1	1	0	0	0	0
6	20	Female	20.4	140	89	286	129	1	1	1	0	0	0	0
7	39	Male	24.2	124	81	292	157	1	1	1	0	0	0	0
8	70	Female	21.9	171	65	170	96	1	0	0	1	0	0	1
9	19	Male	25.2	94	109	201	137	1	1	1	0	0	0	0
10	47	Male	29.2	103	61	246	91	1	0	1	1	0	0	0
11	55	Male	27.1	119	89	200	109	1	0	1	0	0	0	0
12	19	Male	30.5	93	61	232	178	0	0	0	0	0	1	0
13	81	Female	33.3	128	91	204	143	1	1	1	1	0	1	0
14	77	Male	25.2	173	101	226	135	0	0	1	0	0	0	0
15	38	Male	24.6	153	75	276	75	0	1	0	0	0	0	0
16	50	Female	31.8	114	71	242	181	0	0	1	0	0	1	0
17	75	Male	32.2	117	60	237	87	0	1	1	0	0	0	0
18	39	Male	30.7	165	112	229	150	0	1	1	0	0	1	0
19	66	Male	22.2	155	67	241	171	0	0	1	0	1	0	0
20	76	Female	29.1	167	107	196	179	1	1	0	0	0	0	1
21	59	Male	27.5	111	109	187	99	1	0	0	1	0	0	0
22	77	Female	32.3	120	72	288	191	1	0	0	0	0	1	0
23	32	Male	24.1	112	74	164	131	0	0	1	1	0	0	0
24	79	Male	18.7	137	112	150	189	0	0	0	1	0	0	0
25	79	Female	30.5	109	103	284	176	0	0	0	1	0	1	0
26	64	Male	14.2	127	67	150	145	1	0	1	1	0	0	0
27	79	Male	23.3	144	69	198	78	1	0	0	1	0	0	0
28	68	Female	21.9	157	105	245	108	1	1	0	1	1	0	1

Variables (Features)

- **Age:** The age of the individual (in years).
- **Gender:** The gender of the individual (0: Female; 1: Male).
- **BMI:** Body Mass Index (continuous value).
- **Blood Pressure:**
  - Systolic: Upper value of blood pressure (mmHg).
  - Diastolic: Lower value of blood pressure (mmHg).
- **Cholesterol:** Total cholesterol level (mg/dL).

Variables (Lifestyle & Targets)

- **Glucose Level:** Blood glucose level (mg/dL).
- **Smoking / Alcohol Intake / Physical Activity:** Lifestyle factors (0: No; 1: Yes).
- **Family History:** History of chronic diseases in the family (0: No; 1: Yes).
- **Target Variables (Multi-label):**
  - ✓ **Heart_Disease:** Presence of heart disease (0: No; 1: Yes).
  - ✓ **Diabetes:** Presence of diabetes (0: No; 1: Yes).
  - ✓ **Stroke:** Presence of stroke (0: No; 1: Yes).

Data File Description

- **train_dataset.csv**
  - 1,600 cases (80% of data).
  - Includes 11 feature variables and 3 target labels.
- **test_dataset.csv**
  - 400 cases (20% of data).
  - Used for evaluating the trained model.
  - Provided 1 week before the presentation day.

Your Tasks

- **Objective:** Deliver a machine learning model to simultaneously differentiate patients with or without heart disease, diabetes, and stroke.
- **Methods:** Students must explain how they handled the multi-label nature of the data (e.g., Binary Relevance vs. Classifier Chains).
- **Presentation Structure (5/14 Class):**
  - ✓ **Introduction:** Importance of comorbidity prediction.
  - ✓ **Methods:** Data preprocessing and algorithm selection.
  - ✓ **Results:** Performance metrics (AUC, Hamming Loss, etc.).
  - ✓ **Discussion:** Identifying the most influential clinical features for chronic diseases.